

Wentao Ren

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EDUCATION

Nanyang Technological University (NTU) MSc in Signal Processing and Machine Learning •Major Courses: Machine Vision, Pattern Recognition, Neural Network & Deep Learning, Video Signal Processing	Singapore 08/2025 - 12/2026
Beijing University of Technology (BJUT) Beijing - Dublin International College B.Eng. in Electronic & Information Engineering •Major Courses: C Programming Language (A+), University Physics (A+), Linear Algebra (A), Numerical Methods / Optimization (A), Operating Systems (A-), Wireless Systems (A-), Advanced Circuits (A-), Data Structures and Algorithms (A-), Data Mining and Machine Learning (B+), Verilog (B+), Electromagnetics (B+)	Beijing, China 09/2021 - 06/2025
Minor in Innovation & Entrepreneurship (I&E) •Minor Courses: Innovation & Entrepreneurship Project (98), Big Data & AI (90), Equity & Venture Capital Funds (81)	

PUBLICATIONS AND PATENTS

- [1] **Ren WT**, Ke ZH, Xu QZ, et al. Hard Constraint Projection Enhanced Neural Operator with Hybrid Sampling for Chiplet System Thermal Analysis. *Microelectronics Reliability*, Elsevier, 2026.
DOI: <https://doi.org/10.1016/j.microrel.2026.116207>. (**Published, First Author**)
- [2] Zhao JY, Teng ZH, Fan MH, Ma TR, **Ren WT**, Chen S, Liu WC. BitsMoE: Efficient Spectral Energy-Guided Bit Allocation for MoE LLM Quantization. arXiv:2606.00079. (**Under Review**)
- [3] Teng ZH, Zhao JY, **Ren WT**, Fan MH, Ma TR, Chen S, Liu WC. SlimKV: Joint Token-Feature KV Cache Compression with Reconstruction-Free Beacon Attention. (**Under Review**)
- [4] Li XL, Chen ZA, **Ren WT**, Zhang KR, Li WZ. A Method and Application for Testing the Extinction Coefficient of a Solution. Chinese Invention Patent, CN118706767A, 2024. (**Published, Second Student Inventor**)
- [5] Zhang WL, Li YF, **Ren WT**, Liu B. Artificial Intelligence Technology in Alzheimer's Disease Research. *Intractable & Rare Diseases Research*, 2023. DOI: <https://doi.org/10.5582/irdr.2023.01091>. (**Published, Second Student Author, Cited by 59**)

CONFERENCE

AAAI Conference on Artificial Intelligence (AAAI), 2026. Attendee.

RESEARCH EXPERIENCE

Embodied AI World Model Algorithm Research <i>Research Intern, Supervisor: Prof. Kun Li, Institute for AI Industry Research (AIR), Tsinghua University</i> •Conduct research on embodied world models, focusing on latent-space representation learning, generative simulation of environmental dynamics, and World Action Models (WAMs) for action-conditioned prediction and decision-making. •Develop an autoregressive diffusion-based embodied foundation model integrating scene understanding, future-state prediction, action generation, and policy optimization.	05/2026 - Present
AI-Assisted Analog Integrated Circuit Design <i>Graduation Project, Supervisor: Dr. Yuan Gao, Agency for Science, Technology and Research (A*STAR), Singapore</i> •Improve the AnalogCoder-Pro Agent framework and design a system to optimize the analog integrated circuit design. •Investigated machine learning and reinforcement learning methods for analog circuit optimization.	08/2025 - Present
Efficient Inference and Compression for Multimodal Large Language Models <i>Research Assistant, Supervisor: Prof. Weichen Liu, Nanyang Technological University</i> •Investigated compression and efficient inference methods for multimodal large language models (MLLMs), focusing on KV Cache, prompt compression, and GPU memory scheduling optimization. •Developed and evaluated optimization strategies based on open-source models such as Qwen and DeepSeek. •Related results have been developed into two collaborative papers and submitted for review.	08/2025 - 05/2026
PINN Thermal Model of Chiplet Heterogeneous Integration <i>Research Assistant, Supervisor: Prof. Qinzhi Xu, Institute of Microelectronics, Chinese Academy of Sciences</i> •Developed a thermal analysis framework for 2.5D chiplet systems utilizing Physics-Informed Neural Networks (PINNs). •Optimized thermal modeling framework by designing a Deep Operator Network (DeepONet), benchmarked its performance against conventional numerical simulation methods.	07/2024 - 11/2025

• Achieved a 400–1600× speedup while maintaining a 0.01% relative error; published a first-author paper in *Microelectronics Reliability* by Elsevier.

Research Group of Prof. Wenli Zhang (Chief Scientist of National Key R&D Program) BJUT. 07/2023 - 07/2025
Bionic Hand Project | Undergraduate Graduation Project | National Key R&D Program-related Project

- Developed a real-time human-machine interface for a bionic hand using wearable EMG sensors for biosignal acquisition and feature extraction and implementing low-power Bluetooth (BLE) data transmission.
- Designed and deployed a WeChat mini-program interface and backend database, enabling real-time control of the bionic hand via mobile devices with cloud storage for user data.

Intelligent Agricultural Robot Project | “Internet Plus” Innovation and Entrepreneurship Competition

- Developed a fruit maturity evaluation system based on multispectral analysis to enable non-destructive ripeness detection.
- Collaborated on the design of an agricultural harvesting robot to improve data collection range and fruit picking precision.
- Conducted market analysis to evaluate commercialization potential and scalability in precision agriculture applications.

AI Applications in Alzheimer's Disease Research | Research Assistant

- Conducted a comprehensive review of AI applications in early Alzheimer's disease (AD) diagnosis, focusing on non-invasive and portable detection methods.
- Analyzed the advantages of AI-based approaches over traditional clinical and imaging methods, providing insights for future AI-driven predictive models and neurological disease research.

COURSE PROJECTS

Financial Sentiment Classification with FinBERT Fine-Tuning 01/2026 – 04/2026

- Built a three-class financial sentiment classification task to identify negative, neutral, and positive market signals.
- Implemented a TF-IDF + Logistic Regression baseline and fine-tuned FinBERT for financial text classification.
- Conducted multi-seed experiments with early stopping, achieving 97.55% Accuracy / 96.33% Macro-F1.

Image Histogram Equalization and Enhancement Analysis 01/2026 – 04/2026

- Studied histogram equalization for image enhancement, focusing on brightness, contrast, and local details.
- Implemented GHE, BBHE, and CLAHE on a personal face image with multi-metric evaluation.
- Experimental results showed that GHE improves global contrast, BBHE better preserves brightness, and CLAHE performs better in local detail enhancement and over-enhancement suppression.

Face Recognition: Comparative Analysis of Linear and Nonlinear Discriminant Methods 08/2025 – 12/2025

- Compared Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) for feature dimensionality reduction in face recognition tasks.
- Adopted Gaussian Naive Bayes for nonlinear discrimination and evaluated the accuracy and generalization of linear and nonlinear methods under different subspace dimensions.
- Experimental results showed that an appropriate subspace dimension improves generalization, while the MAP-based linear classifier achieved the most stable performance.

INTER-SCHOOL EXCHANGE

MIT xPRO – Blended AI+X Program 11/2025 - Present

- Participated in a blended AI+X program that combined online courses on AI fundamentals and cross-disciplinary applications with an on-site workshop series and project-based mentoring at Massachusetts Institute of Technology (MIT).

Nanyang Technological University 01/2024 - 02/2024

- Carried out a comprehensive study of Artificial Intelligence, Ethics, Business Strategies, Business Networks of Overseas Entrepreneurs, Multinational Enterprise Management, AI Frontiers and Industry Trends, and Public Policy.

Imperial College London 01/2023 - 02/2023

- Performed a holistic study of how to innovate robotics, the Internet of Things (IoT), and Artificial Intelligence (AI).

AWARDS/HONORS

Outstanding Undergraduate Graduation Design Award at BJUT	2025
Outstanding Graduate Award	2025
Academic Excellence Award and Innovation Award.	2022 - 2025
Outstanding Student Leader Award at BJUT	2022 & 2025
Honorable Mention, Mathematical Contest in Modeling (MCM)	2023

SKILLS

- **Simulation Software:** LTspice, Logisim, MATLAB, COMSOL, Photoshop, Lightroom, DaVinci, Overleaf, Jupyter.
- **Programming & Operating Systems:** Proficient in C, Python (Pytorch), Verilog, Latex, Linux.
- **Languages:** English (IELTS 6.5)